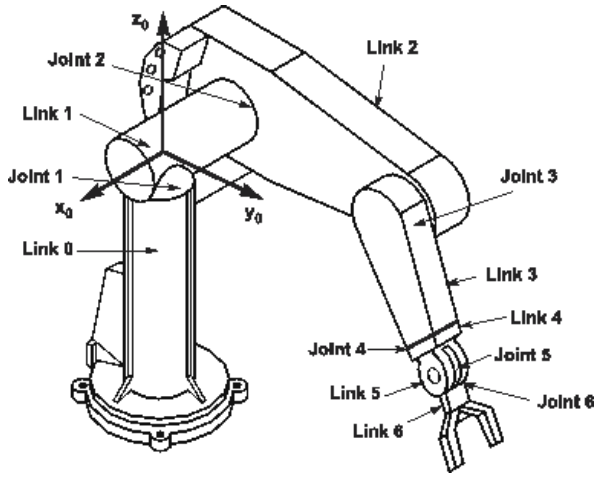
Lecture 4: Homogenous Transformations

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What did we learn last time?

Recap of Lecture 5

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What are we doing today?

1. Introduction to Robotics & Recap of Linear Algebra and Mathematical Notation
2. Translations & Rotation Matrices
3. Other Representations for Orientation
4. **Transformation Matrices** ***(Today)***
5. Forward Kinematics & DH Parameters
6. Schematic Representations of Robots, FK from DH, DH Parameters Practice
7. Inverse Kinematics
8. Kinematic Simulation
9. Angular Velocities, Velocity Propagation & Trajectory Generation
10. The Jacobian Matrix, Singularities
11. Analytical & Geometrical Jacobian, Numerical IK, Statics
12. Manipulability, More on the Robotic Systems Toolbox

Topics for Today

Part I:

- Short **recap** of **translation vectors** and **rotation matrices**
- **Mappings** between frames (**combining translation** and **rotation**)

Part II

- From Cartesian to **Homogenous Coordinates**
- **Homogenous Transformation Matrices**
  - How to **construct them**
  - **Properties**
  - Some **computational considerations**
  - A note on **notation** (pun possibly intended)
  - **Visualizing** chains of transformations

Part III

- **Practical Applications**

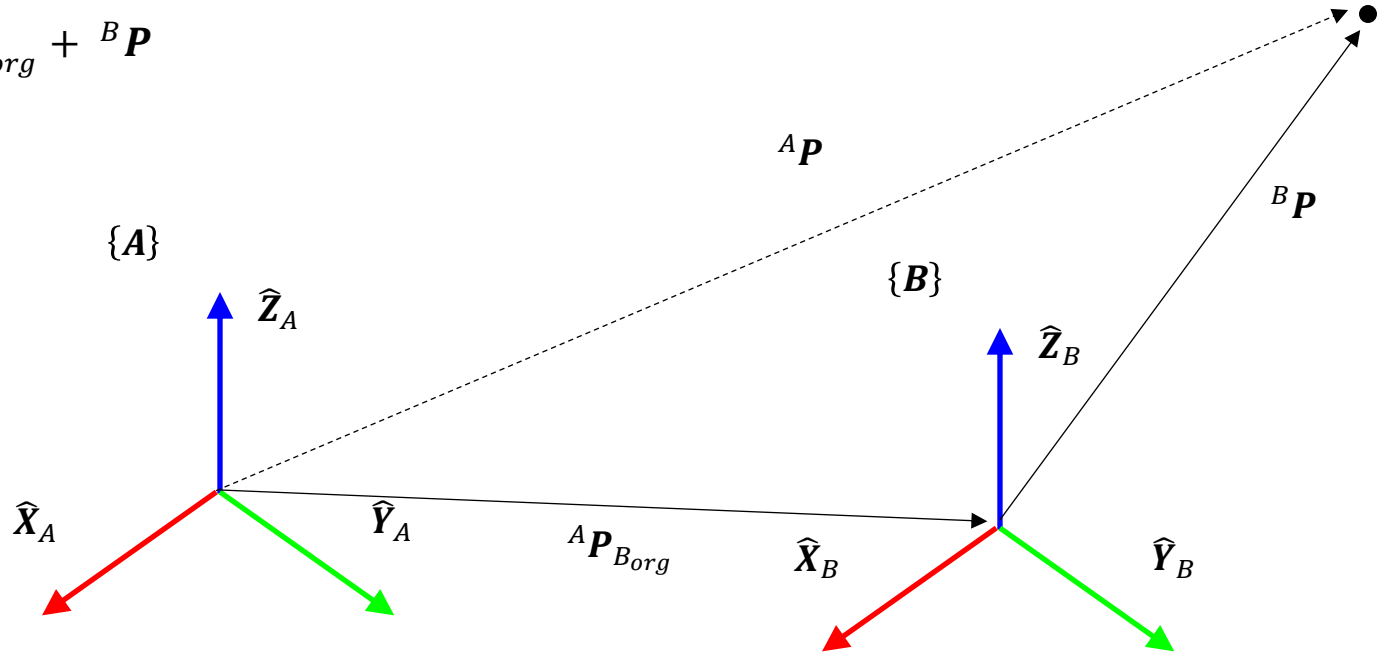
Part I: Mappings between Frames

with Translation Vectors and Rotation Matrices

Translational Mappings (with Vectors)

Vectors can be used to represent **points** and **translations** between frames:

$$^{A}\boldsymbol{P} = {}^{A}\boldsymbol{P}_{B_{org}} + {}^{B}\boldsymbol{P}$$

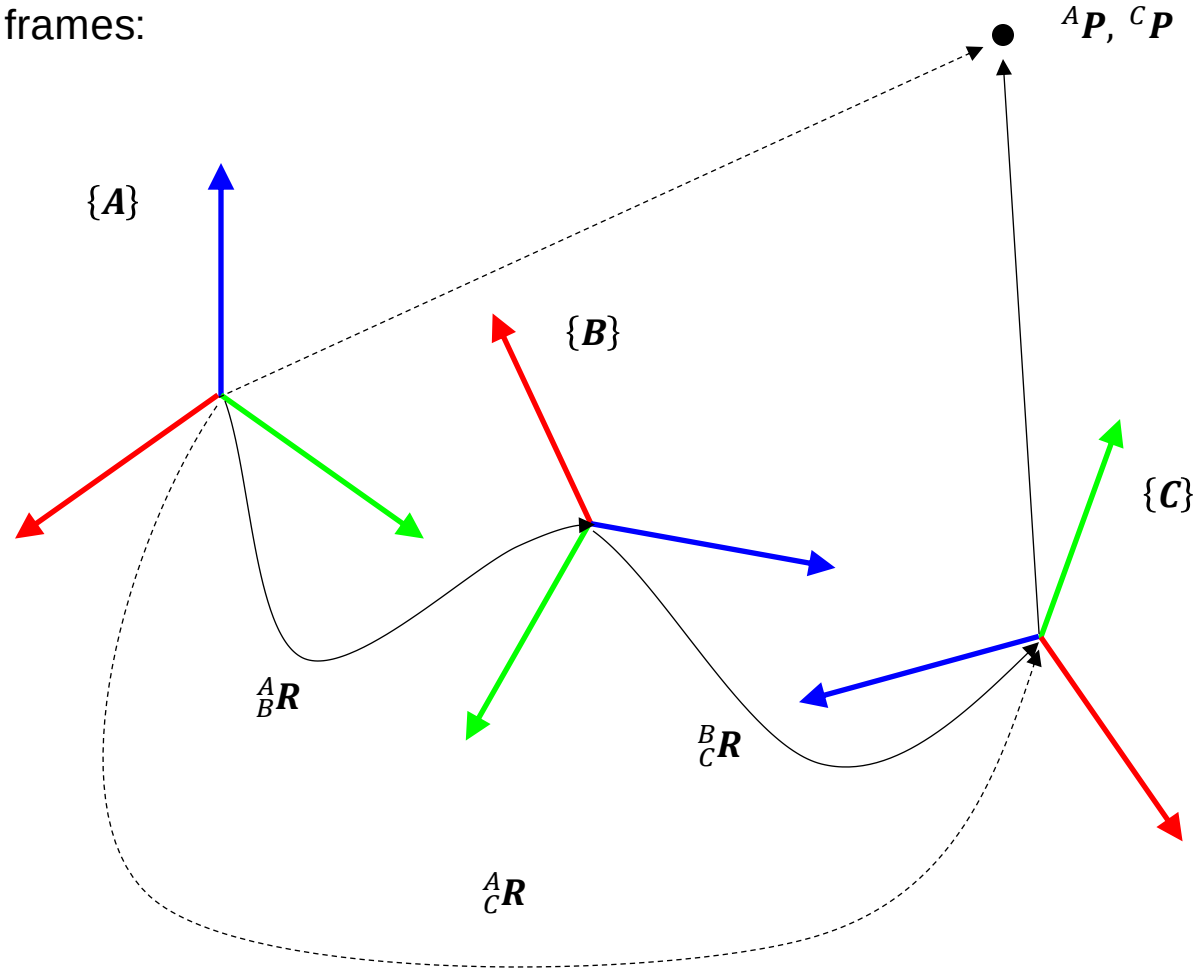


Rotational Mappings (with Rotation Matrices)

Matrices can be used to represent **rotations** between frames:

$$ {}^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\ {}^{B}_{C}\boldsymbol{R}\ {}^{C}\boldsymbol{P} = {}^{A}_{C}\boldsymbol{R}\ {}^{C}\boldsymbol{P} $$

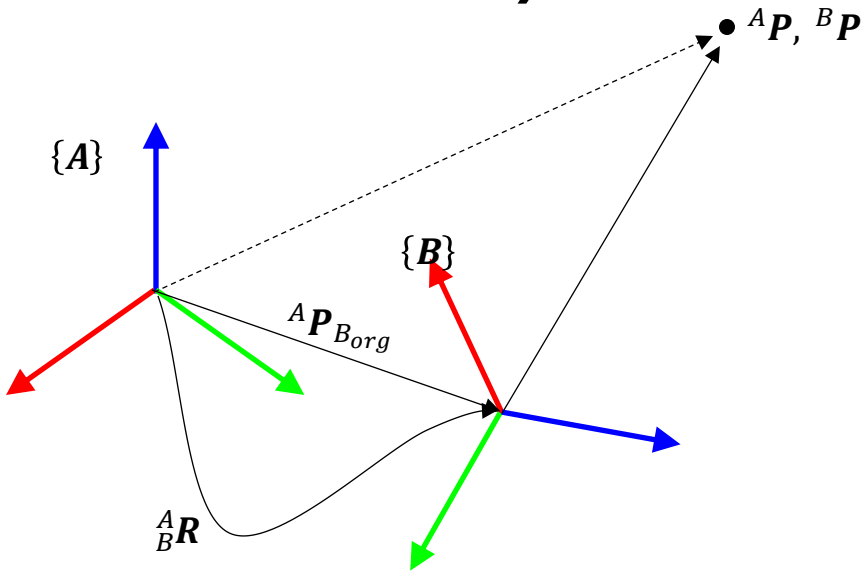
$$ {}^{C}\boldsymbol{P} = {}^{B}_{C}\boldsymbol{R}^{-1}\ {}^{A}_{B}\boldsymbol{R}^{-1}\ {}^{A}\boldsymbol{P} = {}^{A}_{C}\boldsymbol{R}^{-1}{}^{A}\boldsymbol{P} $$



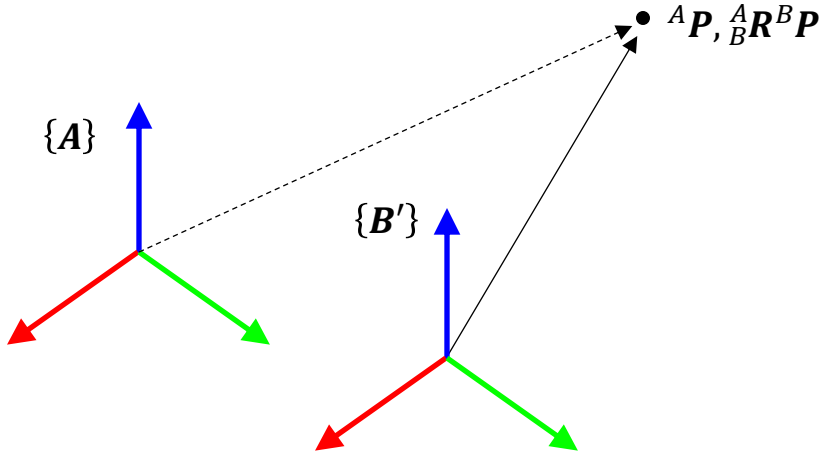
General Mappings (with both Translation & Rotation)

Apply **rotation first**, **then translation:**

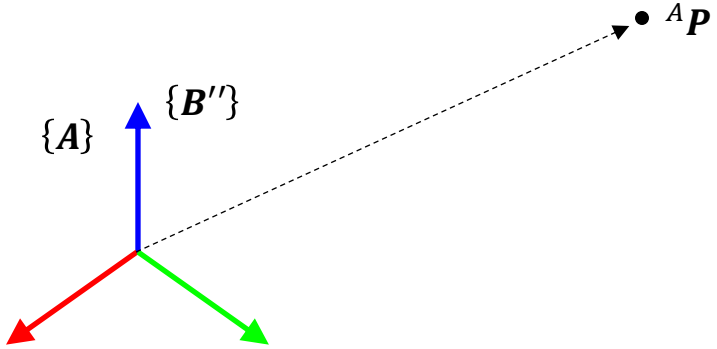
$$^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\,{}^{B}\boldsymbol{P} + {}^{A}\boldsymbol{P}_{B_{org}}$$



1. After applying rotation $\left({}^{A}_{B}\boldsymbol{R}\,{}^{B}\boldsymbol{P}\right)$:



2. After applying translation $\left({}^{A}\boldsymbol{P}_{B_{org}}\right)$:



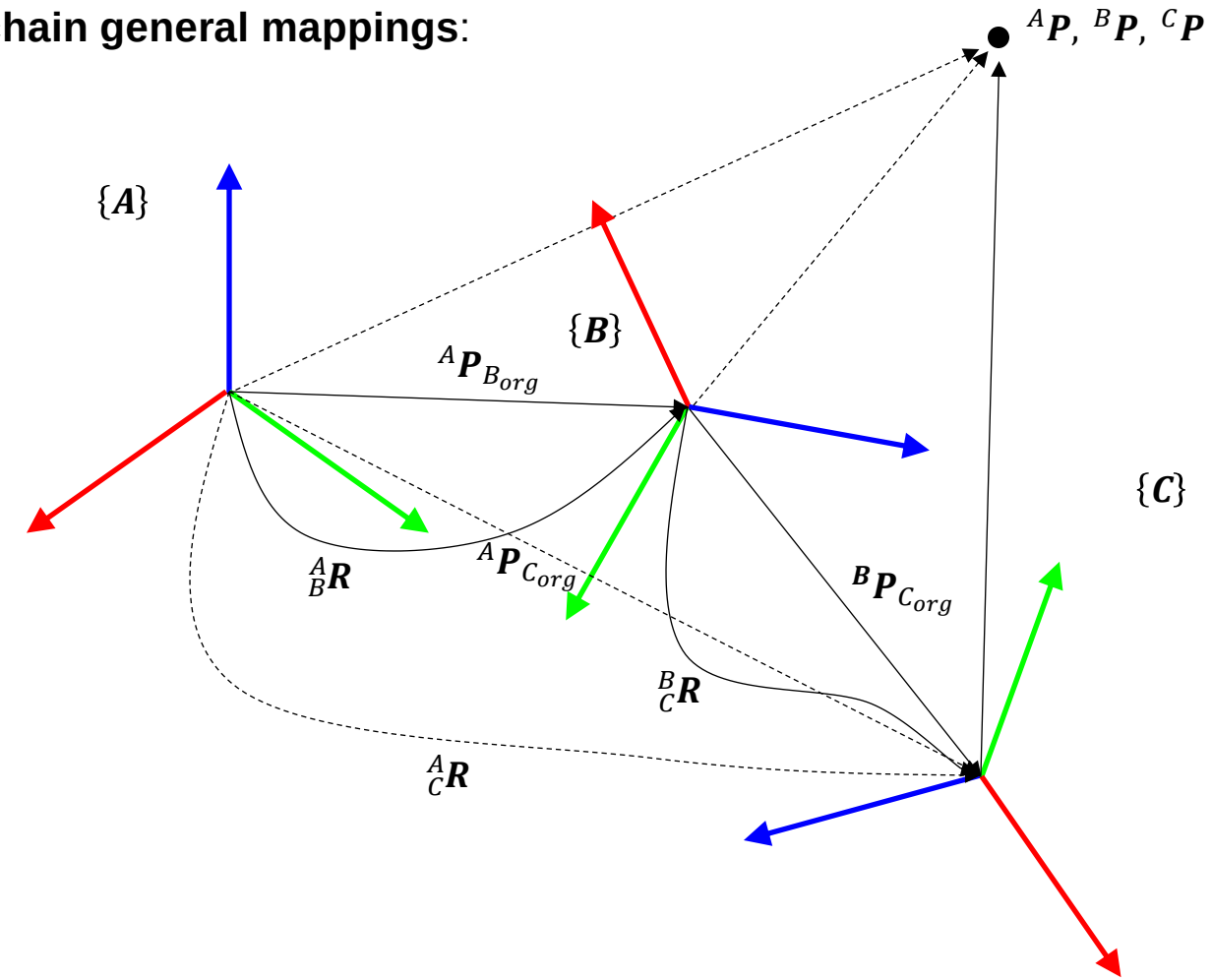
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General Mappings (in Multiple Frames)

As with pure rotations and pure translations, **we can chain general mappings**:

$$^{B}\boldsymbol{P} = {}^{B}_{C}\boldsymbol{R}\ {}^{C}\boldsymbol{P} + {}^{\boldsymbol{B}}\boldsymbol{P}_{C_{org}}$$
$$^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\ {}^{B}\boldsymbol{P} + {}^{\boldsymbol{A}}\boldsymbol{P}_{B_{org}}$$

$$^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\left({}^{B}_{C}\boldsymbol{R}\ {}^{C}\boldsymbol{P} + {}^{\boldsymbol{B}}\boldsymbol{P}_{C_{org}}\right) + {}^{\boldsymbol{A}}\boldsymbol{P}_{B_{org}}$$



The Grass is Always Greener on the Other Side…

So… we have:

- **Vectors** to represent **points** and **translations**
- **Matrices** (among others) to represent **orientations** and **rotations**

…and we can perform **mappings** by applying **rotation followed by translation.**

Why aren't we happy?



$$^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\left({}^{B}_{C}\boldsymbol{R}\,{}^{C}\boldsymbol{P} + {}^{B}\boldsymbol{P}_{C_{org}}\right) + {}^{A}\boldsymbol{P}_{B_{org}}$$

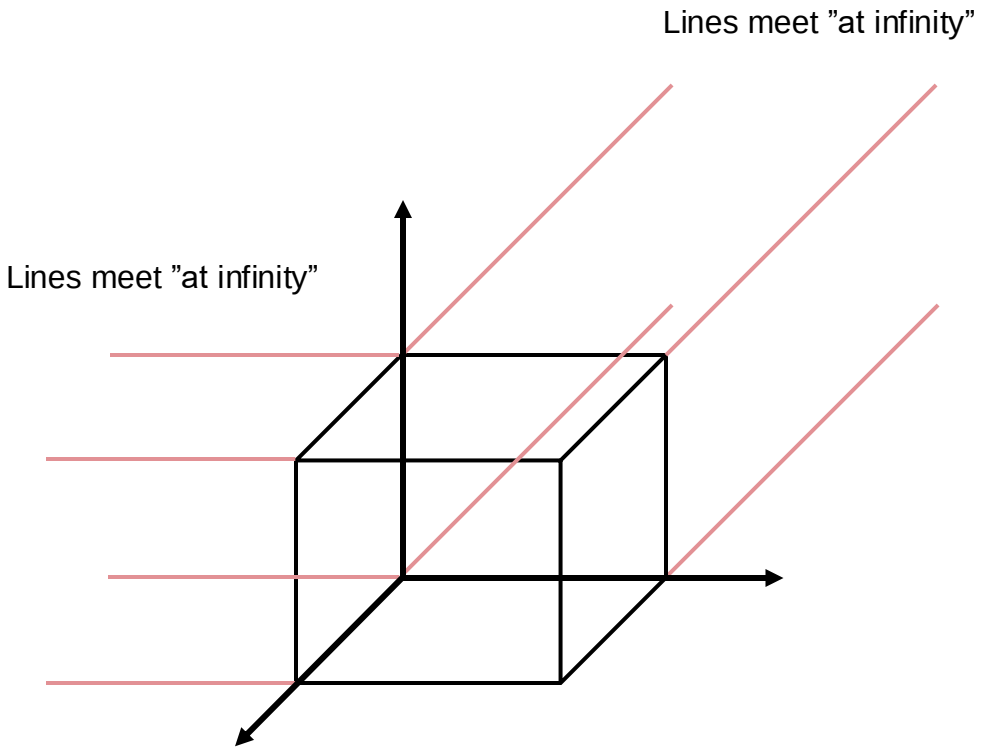
$$^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{T}\,{}^{B}_{C}\boldsymbol{T}\,{}^{C}\boldsymbol{P}$$

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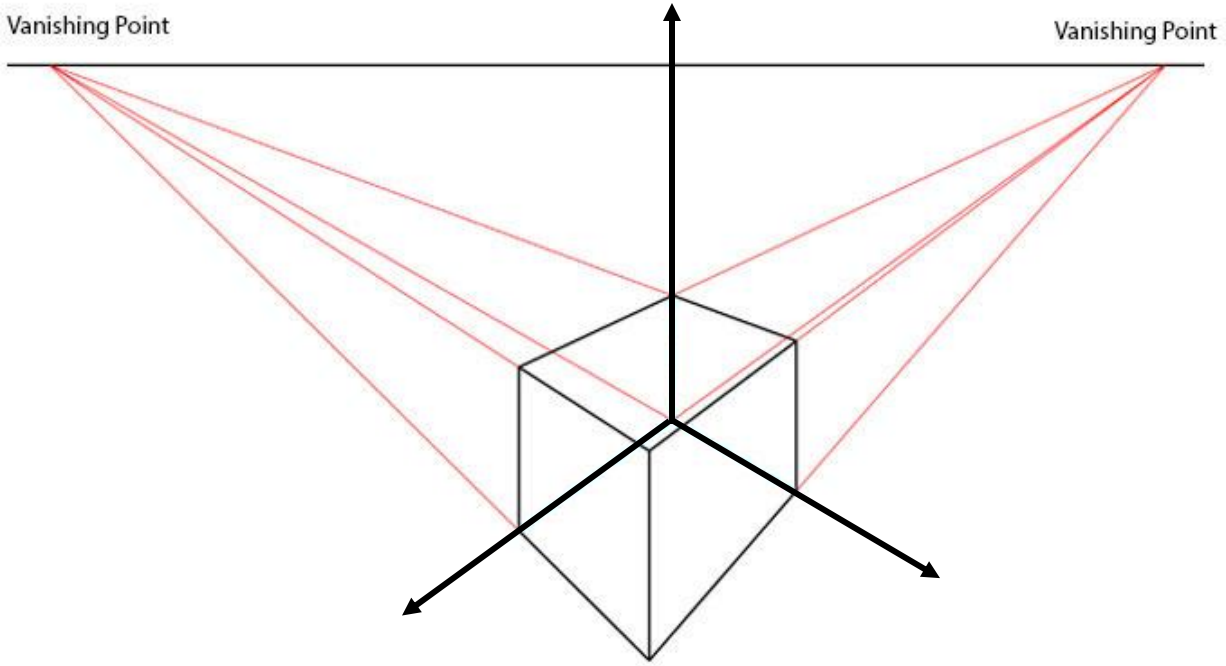
Part II: Homogenous Transformation Matrices

From Euclidean to Projective Geometry

Euclidean Geometry
(Cartesian coordinates)



Projective Geometry
(Homogenous coordinates)



Source: https://brilliant.org/wiki/projective-geometry/

From Cartesian to Homogenous Coordinates

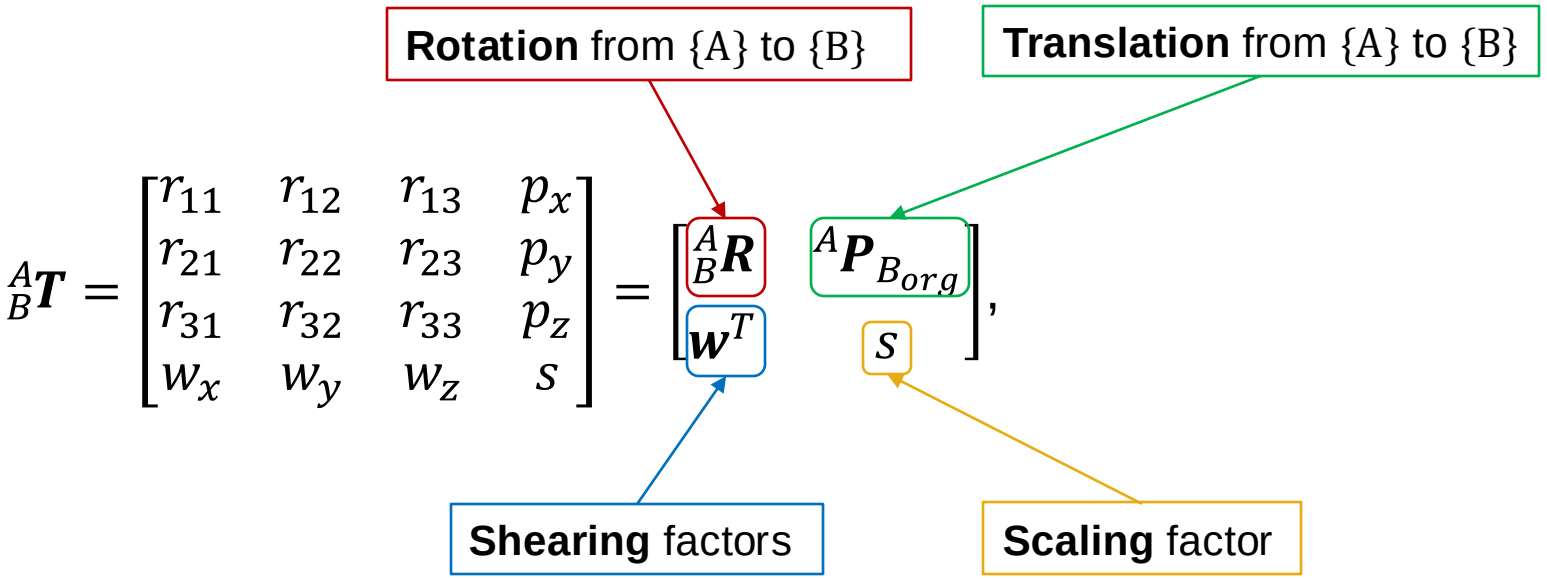
To **convert** from **Cartesian to Homogenous** coordinates:

1. Add an **extra row** to the vector.
2. The element of this row is a **number, *s***.
3. **Multiply** every other element **by *s***.

	Cartesian		**Homogenous**
2D:	$^{A}\boldsymbol{P}_{B_{org}} = \begin{bmatrix} x \\ y \end{bmatrix}$	⟶	$^{A}\boldsymbol{P}_{B_{org}} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \\ 2 \end{bmatrix} = \begin{bmatrix} s \cdot x \\ s \cdot y \\ s \end{bmatrix}$
3D:	$^{A}\boldsymbol{P}_{B_{org}} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$	⟶	$^{A}\boldsymbol{P}_{B_{org}} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \\ 2z \\ 2 \end{bmatrix} = \begin{bmatrix} s \cdot x \\ s \cdot y \\ s \cdot z \\ s \end{bmatrix}$

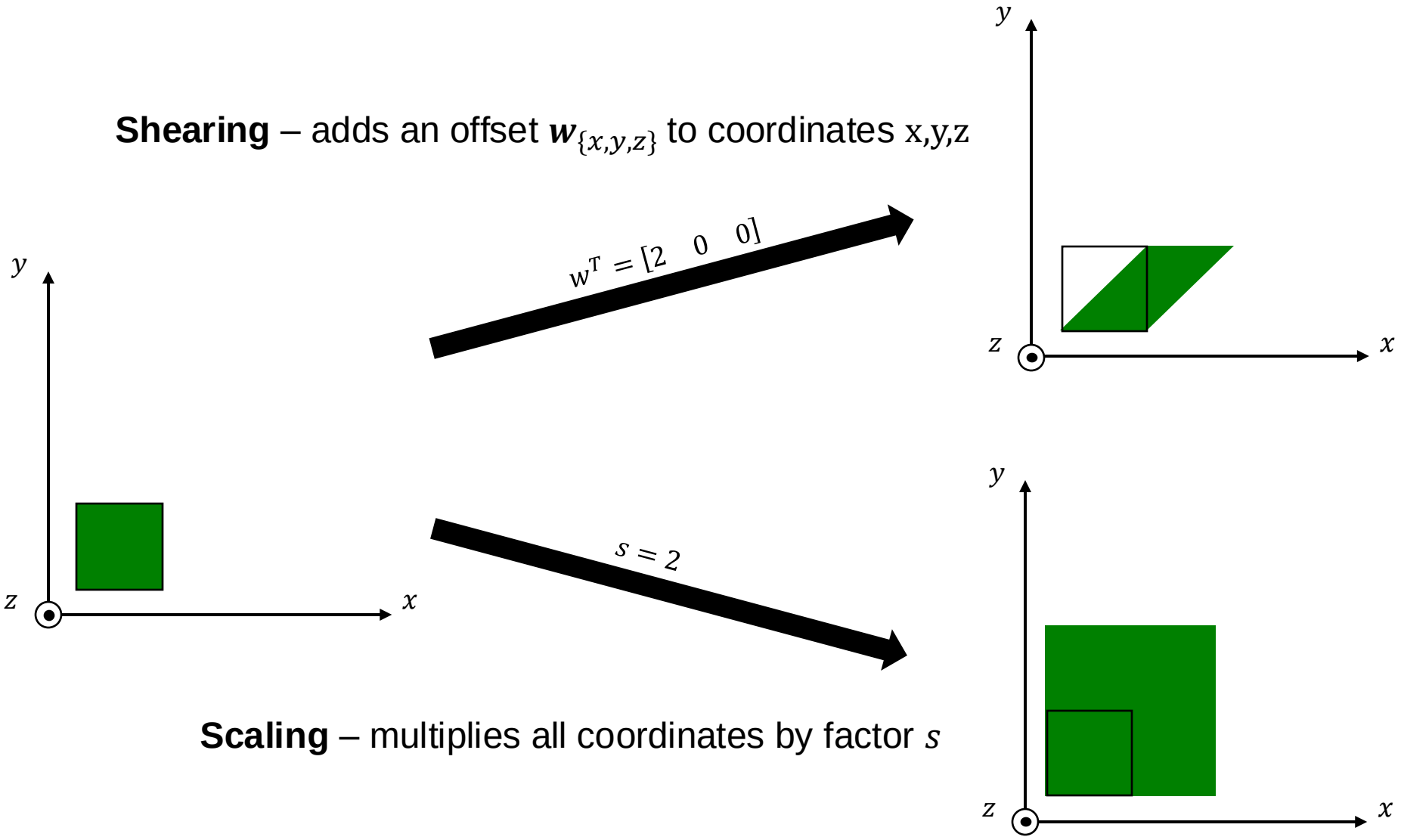
Generic Homogenous Transformations

The reason for using **homogenous coordinates** is that they allow us **combine rotation and translation**:



Shearing and Scaling

Shearing – adds an offset $\boldsymbol{w}_{\{x,y,z\}}$ to coordinates x,y,z



Scaling – multiplies all coordinates by factor s

Homogenous Transformations
in Robot Kinematics (shearing and scaling disabled)

In robot kinematics, we are **only interested in combining rotation and translation***, so:

- **Shearing** is **disabled,**
- **Scaling** is set to **1**.

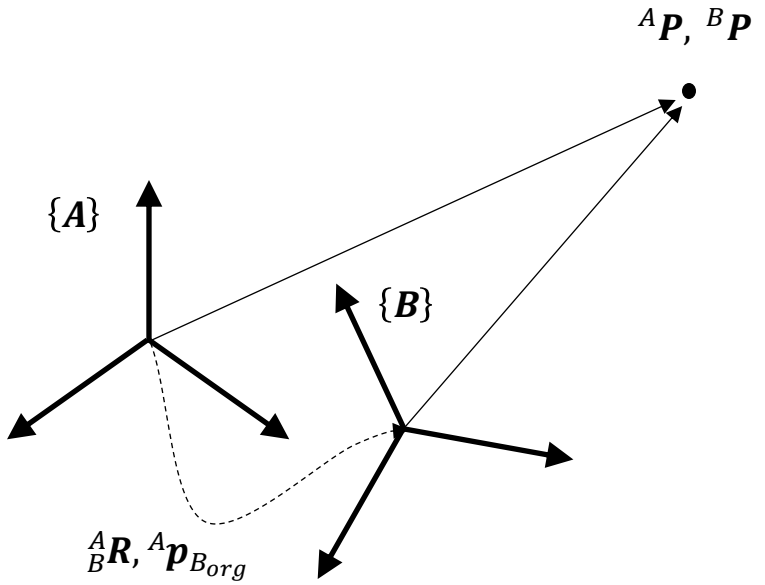
$$
{}^{A}_{B}\boldsymbol{T} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}\boldsymbol{R} & {}^{A}\boldsymbol{p}_{B_{org}} \\ \boldsymbol{0}^{3\times 1} & 1 \end{bmatrix}
$$

Disabled shearing/scaling

**In Computer Vision, however, shearing and scaling are very useful, as you will see in future courses.*

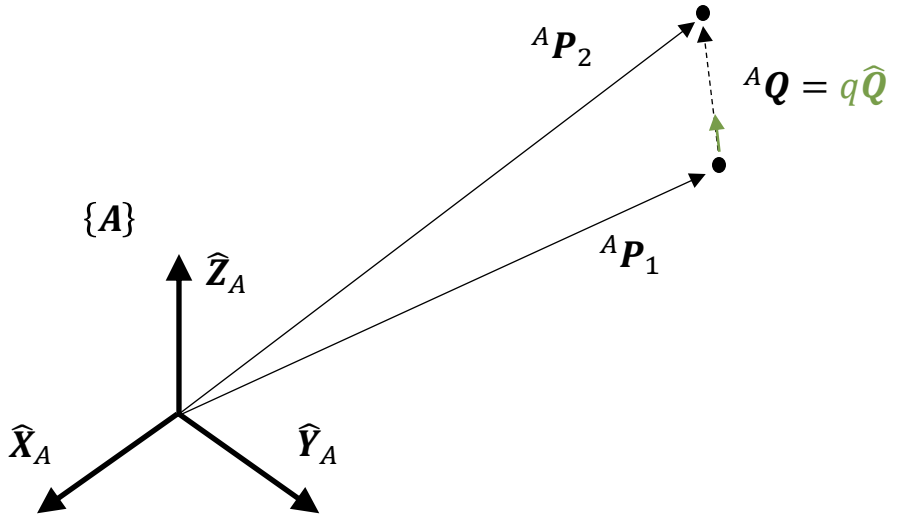
Homogenous Transformations on Points

$$
\begin{aligned}
{}^{A}\boldsymbol{P} &= {}^{A}_{B}\boldsymbol{T}\ {}^{B}\boldsymbol{P} \\
\Rightarrow \begin{bmatrix} {}^{A}\boldsymbol{P} \\ 1 \end{bmatrix} &= \begin{bmatrix} {}^{A}_{B}\boldsymbol{R} & {}^{A}\boldsymbol{p}_{B_{org}} \\ \boldsymbol{0}^{3\times 1} & 1 \end{bmatrix} \begin{bmatrix} {}^{B}\boldsymbol{P} \\ 1 \end{bmatrix}
\end{aligned}
$$



Homogenous Transformations: Pure Translation

$$\boldsymbol{D_Q}(q) = \begin{bmatrix} 1 & 0 & 0 & q_x \\ 0 & 1 & 0 & q_y \\ 0 & 0 & 1 & q_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{I}^{3\times3} & q\widehat{\boldsymbol{Q}} \\ \boldsymbol{0}^{3\times1} & 1 \end{bmatrix}$$



Translation

$({}^A\boldsymbol{P}_2 = {}^A\boldsymbol{P}_1 + {}^A\boldsymbol{Q})$

$\Longrightarrow {}^A\boldsymbol{P}_2 = \boldsymbol{D_Q}(q)\,{}^A\boldsymbol{P}_1$

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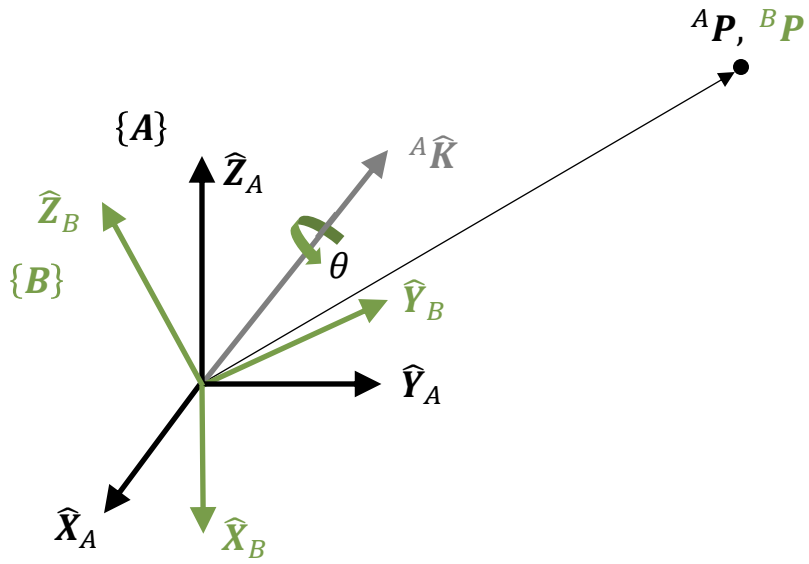
Homogenous Transformations: Pure Rotation

Rotation

$({}^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\,{}^{B}\boldsymbol{P})$

$$\boldsymbol{R_K}(\theta) = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}\boldsymbol{R} & \boldsymbol{0}^{1\times 3} \\ \boldsymbol{0}^{3\times 1} & 1 \end{bmatrix}$$

$$\Longrightarrow {}^{A}\boldsymbol{P} = \boldsymbol{R_K}(\theta)\,{}^{B}\boldsymbol{P}$$



Computational Considerations

When writing **code**, we typically want it to be **as efficient as possible**.

A transformation matrix **always** has the following structure: ${}^{A}_{B}\boldsymbol{T} = \begin{bmatrix} {}^{A}_{B}\boldsymbol{R} & {}^{A}\boldsymbol{p}_{B_{org}} \\ \boldsymbol{0}^{3\times1} & 1 \end{bmatrix}$

Exercise:

In small groups:

1. Can you think of a way to **optimize the storage space** for a transformation matrix in computer memory?
2. Computationally, finding the **inverse** of a matrix is **very expensive**, but taking the **transpose** is **cheap**.

   Can you think of an **efficient** way to find the **inverse** of a **transformation matrix**?

   (i.e. if we know ${}^{A}_{B}\boldsymbol{T}$, how can we efficiently compute ${}^{B}_{A}\boldsymbol{T} = {}^{A}_{B}\boldsymbol{T}^{-1}$?)

   Hint 1: Make a drawing of the chain of transformations.

   Hint 2: A transformation can be written as ${}^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\,{}^{B}\boldsymbol{P} + {}^{A}\boldsymbol{P}_{B_{org}}$ *. What do we need to find its inverse?*

   Hint 3: What were the special properties of the rotation matrix? Can we use them here?

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Exercise Solution (Efficient Inverse)

1. We can just store it as a 3x4 matrix, since the last row is always $[0 \quad 0 \quad 0 \quad 1]$.
2. We know: ${}^{A}_{B}\boldsymbol{R}$, ${}^{A}\boldsymbol{P}_{B_{org}}$ (from decomposing ${}^{A}_{B}\boldsymbol{T}$)

For any point we can write: ${}^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\,{}^{B}\boldsymbol{P} + {}^{A}\boldsymbol{P}_{B_{org}}$ and ${}^{B}\boldsymbol{P} = {}^{B}_{A}\boldsymbol{R}\,{}^{A}\boldsymbol{P} + {}^{B}\boldsymbol{P}_{A_{org}}$

We **don't know**: ${}^{B}\boldsymbol{P}_{A_{org}}$

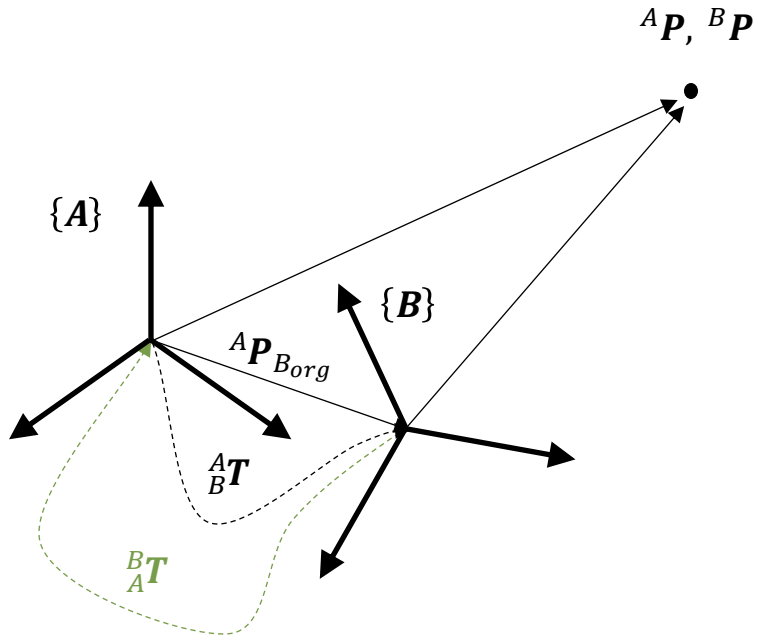
We can visualize ${}^{A}\boldsymbol{P}_{B_{org}}$ as if seen from {B}:

$$
\begin{aligned}
{}^{B}\boldsymbol{P} &= {}^{B}_{A}\boldsymbol{R}\,{}^{A}\boldsymbol{P} + {}^{B}\boldsymbol{P}_{A_{org}} \\
{}^{B}\left({}^{A}\boldsymbol{P}_{B_{org}}\right) &= {}^{B}_{A}\boldsymbol{R}\left({}^{A}\boldsymbol{P}_{B_{org}}\right) + {}^{B}\boldsymbol{P}_{A_{org}} = 0 \\
{}^{B}\boldsymbol{P}_{A_{org}} &= -{}^{B}_{A}\boldsymbol{R}\left({}^{A}\boldsymbol{P}_{B_{org}}\right)
\end{aligned}
$$

We know ${}^{B}_{A}\boldsymbol{R} = {}^{A}_{B}\boldsymbol{R}^{T}$ because rotation matrices are **orthogonal.**

Therefore, the **inverse of a transformation** can be **written as**:

$$
{}^{A}_{B}\boldsymbol{T}^{-1} = {}^{B}_{A}\boldsymbol{T} = \begin{bmatrix} ({}^{A}_{B}\boldsymbol{R}^{T}) & -({}^{A}_{B}\boldsymbol{R}^{T})\,{}^{A}\boldsymbol{P}_{B_{org}} \\ \boldsymbol{0}^{3\times 1} & 1 \end{bmatrix}
$$



Remember this?



There is no such thing as
The Golden Notation™ in Robotics.

Craig's notation is not the only notation.

(We will see some other examples later in the course.)

A Note on Notation (pun somewhat intended)

There are **many ways to denote frames and transformations**.

Craig's notation is arguably **not very common**.

	Craig	"Standard"	"Alternate"
Point in frame {A}	${}^{A}\boldsymbol{P}$	$\boldsymbol{p}^{A}$	$\boldsymbol{p}_{A}$
Rotation from {A} to {B} (or orientation of {B} as seen from {A})	${}^{A}_{B}\boldsymbol{R}$	$\boldsymbol{R}^{A}_{B}$	$\boldsymbol{R}^{B}_{A}$
Transformation from {A} to {B} (or position + orientation of {B} as seen from {A})	${}^{A}_{B}\boldsymbol{T}$	$\boldsymbol{T}^{A}_{B}$	$\boldsymbol{T}^{B}_{A}$
Main Features	• Both **vectors** and **matrices** in **capital bold letters**. • **Left-indexing** • **"From"** on **top**, **"to"** on **bottom**.	• **Vectors** in **small bold letters**. • **Matrices** in **capital bold letters**. • **Right-indexing** • **"From"** on **top**, **"to"** on **bottom**.	• **Same as standard** • **"To"** on **top**, **"From"** on **bottom**.

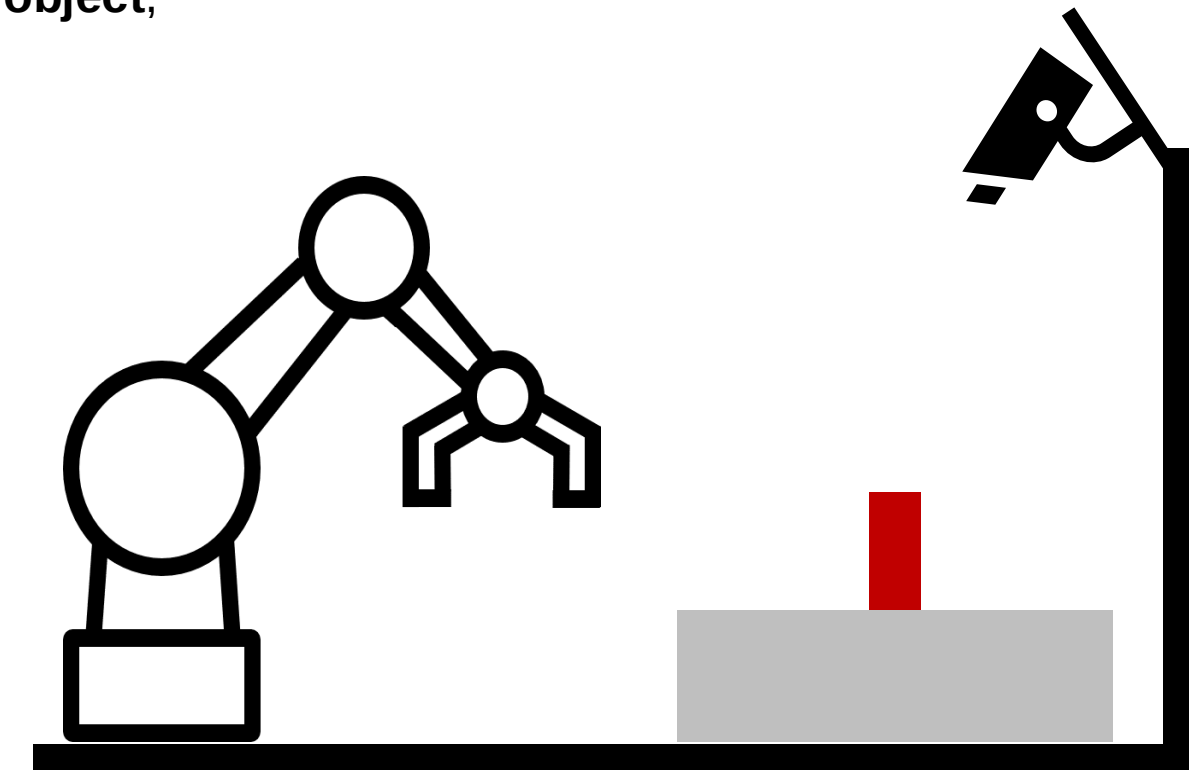
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Part III: Practical Applications



Example 1: Industrial Robot

The robot needs to **move** in order to **grasp** the **red object**,
Whose **position** is **determined by a camera**.



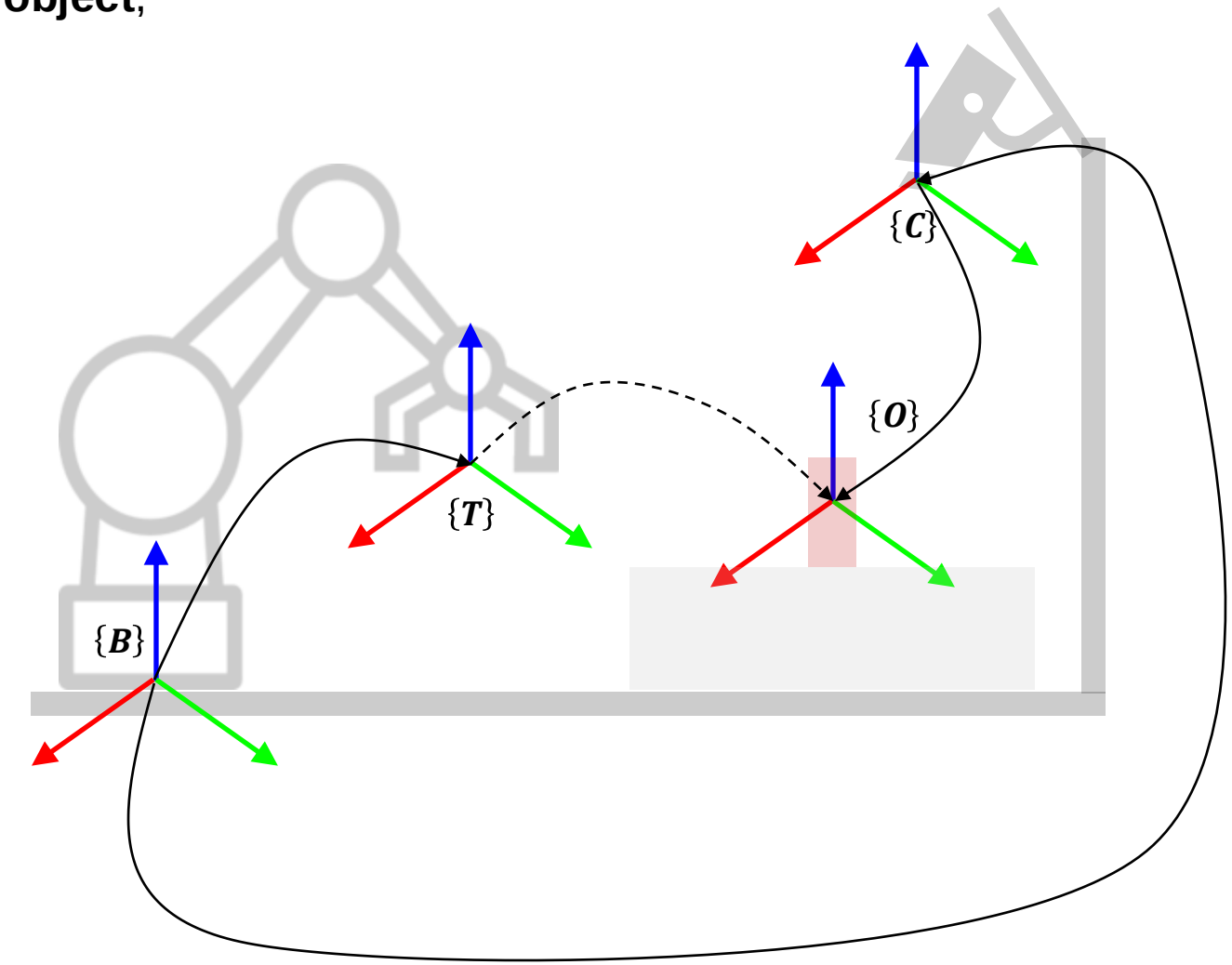
Example 1: Industrial Robot

The robot needs to **move** in order to **grasp** the **red object**,
Whose **position** is **determined by a camera**.

Frames:
$\{\mathbf{B}\}$: Robot base
$\{\mathbf{T}\}$: Robot tool (TCP)
$\{\mathbf{C}\}$: Camera
$\{\mathbf{O}\}$: Object

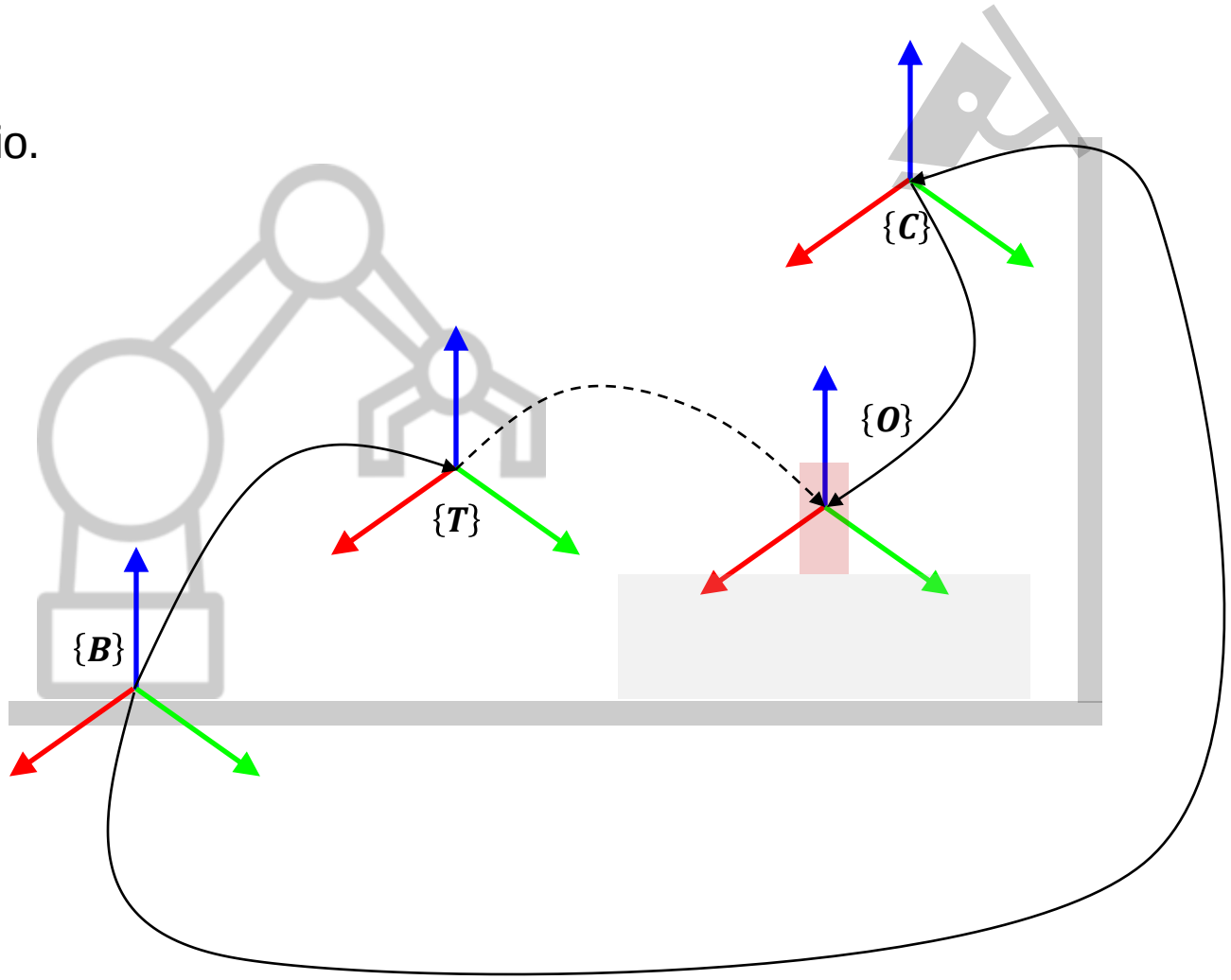
Known:
$\{\mathbf{B}\}$: by definition
${}^{B}_{T}\boldsymbol{T}$: forward kinematics
${}^{B}_{C}\boldsymbol{T}$, ${}^{C}_{O}\boldsymbol{T}$: from measurements/calibration

Wanted: ${}^{T}_{O}\boldsymbol{T}$



Example 1: Industrial Robot

1. Write down the transform equation for the scenario.
2. Solve the equation for ${}^{T}_{O}\boldsymbol{T}$.



Example 1: Industrial Robot – Two Solutions

Think of it **mathematically**:

$$ {}^{B}_{T}\boldsymbol{T}\,{}^{T}_{O}\boldsymbol{T} = {}^{B}_{C}\boldsymbol{T}\,{}^{C}_{O}\boldsymbol{T} $$

$$ {}^{B}_{T}\boldsymbol{T}^{-1}\,{}^{B}_{T}\boldsymbol{T}\,{}^{T}_{O}\boldsymbol{T} = {}^{B}_{T}\boldsymbol{T}^{-1}\,{}^{B}_{C}\boldsymbol{T}\,{}^{C}_{O}\boldsymbol{T} $$

$$ {}^{T}_{O}\boldsymbol{T} = {}^{B}_{T}\boldsymbol{T}^{-1}\,{}^{B}_{C}\boldsymbol{T}\,{}^{C}_{O}\boldsymbol{T} $$

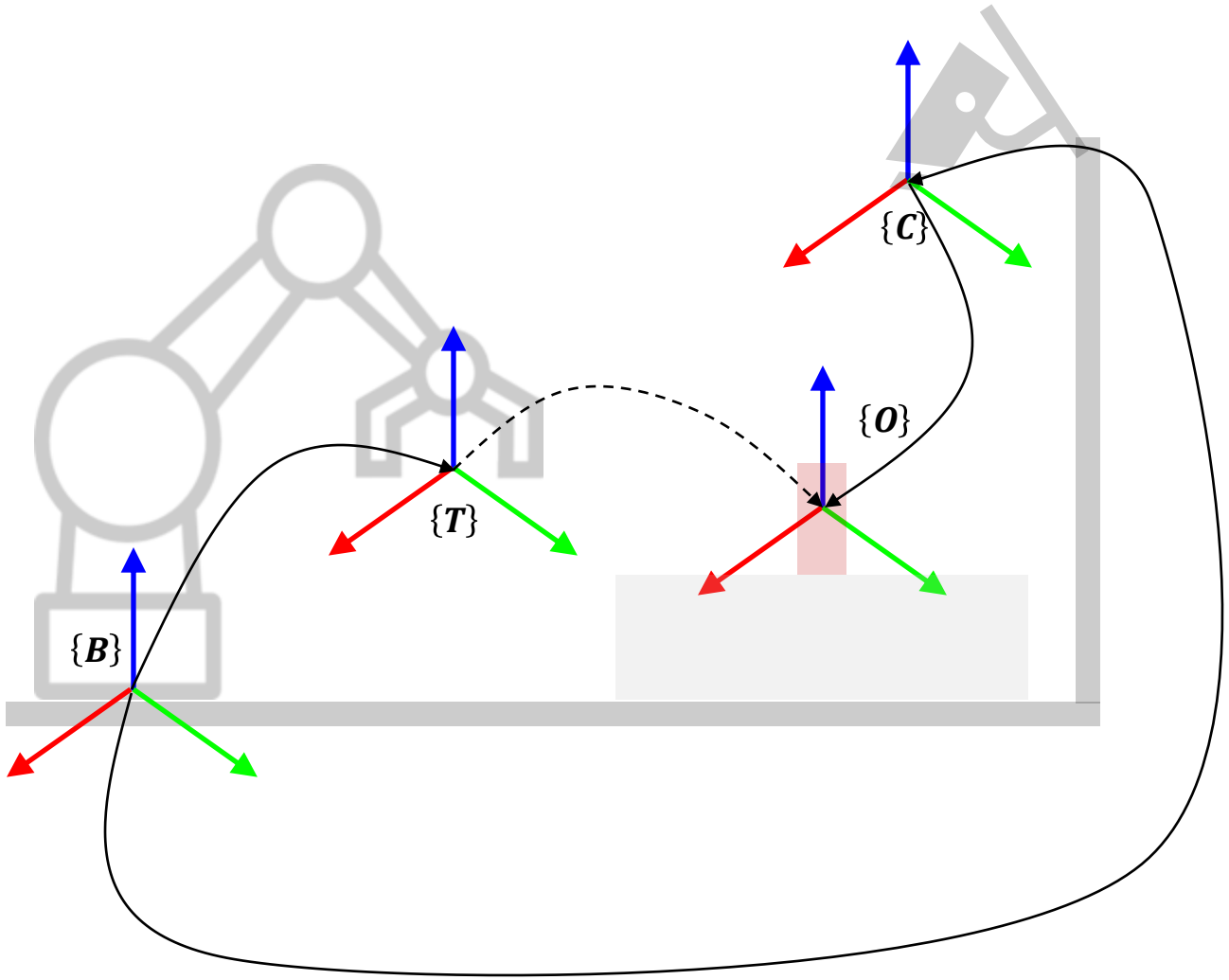
Think of it **graphically**:

You want to get from $\{\boldsymbol{T}\}$ to $\{\boldsymbol{O}\}$, following the arrows.

1. The first arrow goes from $\{\boldsymbol{B}\}$ to $\{\boldsymbol{T}\}$, so we need to reverse it. → ${}^{B}_{T}\boldsymbol{T}^{-1}$.

2. The rest of the arrows point in the right direction, so we just follow them. → ${}^{B}_{C}\boldsymbol{T}\,{}^{C}_{O}\boldsymbol{T}$.

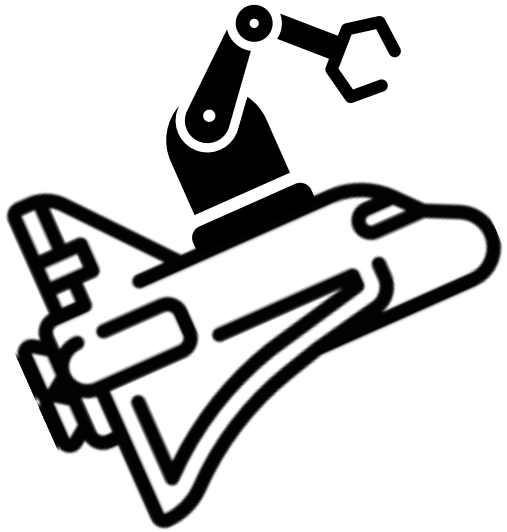
3. Multiply it all together. → ${}^{T}_{O}\boldsymbol{T} = {}^{B}_{T}\boldsymbol{T}^{-1}\,{}^{B}_{C}\boldsymbol{T}\,{}^{C}_{O}\boldsymbol{T}$

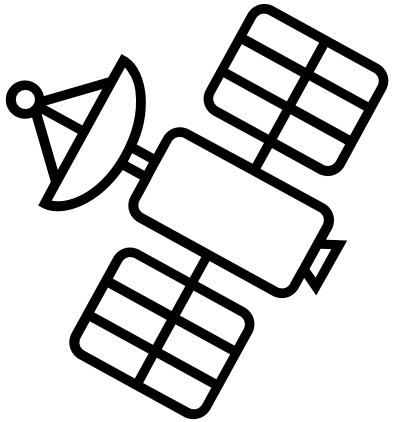


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Example 2: Space Robotics

The shuttle-mounted robot needs a trajectory to repair a satellite's antenna in space.







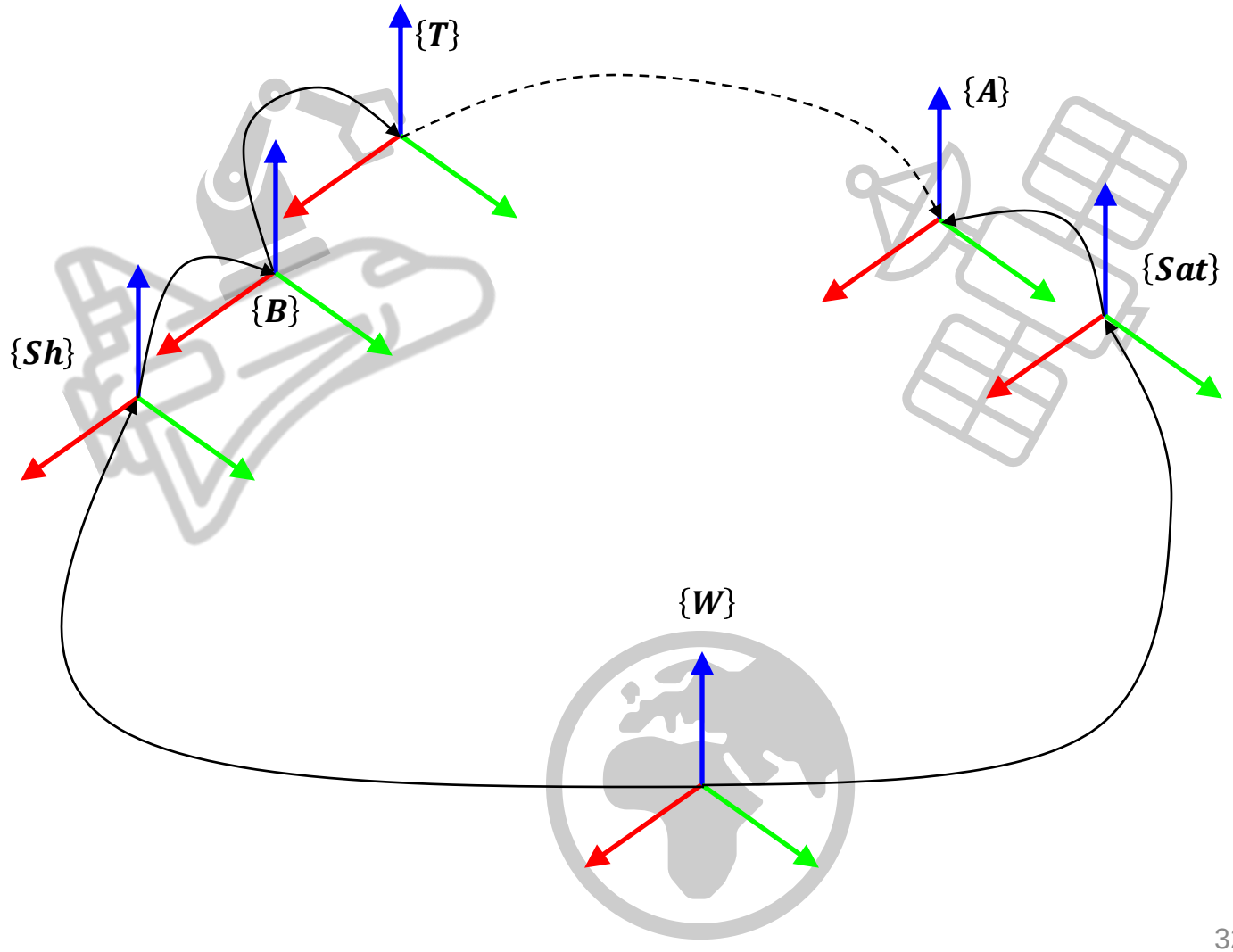
Example 2: Space Robotics

The shuttle-mounted robot needs a trajectory to repair a satellite's antenna in space.

Frames:
$\{\mathbf{W}\}$: World coordinates
$\{\mathbf{B}\}$: Robot base
$\{\mathbf{T}\}$: Robot tool (TCP)
$\{\mathbf{Sh}\}$: Shuttle
$\{\mathbf{Sat}\}$: Satellite
$\{\mathbf{A}\}$: Antenna

Known:
$\{\mathbf{W}\}$: by definition
${}^{B}_{T}\boldsymbol{T}$: forward kinematics
${}^{W}_{Sh}\boldsymbol{T}$, ${}^{W}_{Sat}\boldsymbol{T}$: from measurements
${}^{Sh}_{B}\boldsymbol{T}$, ${}^{Sat}_{A}\boldsymbol{T}$: from measurements

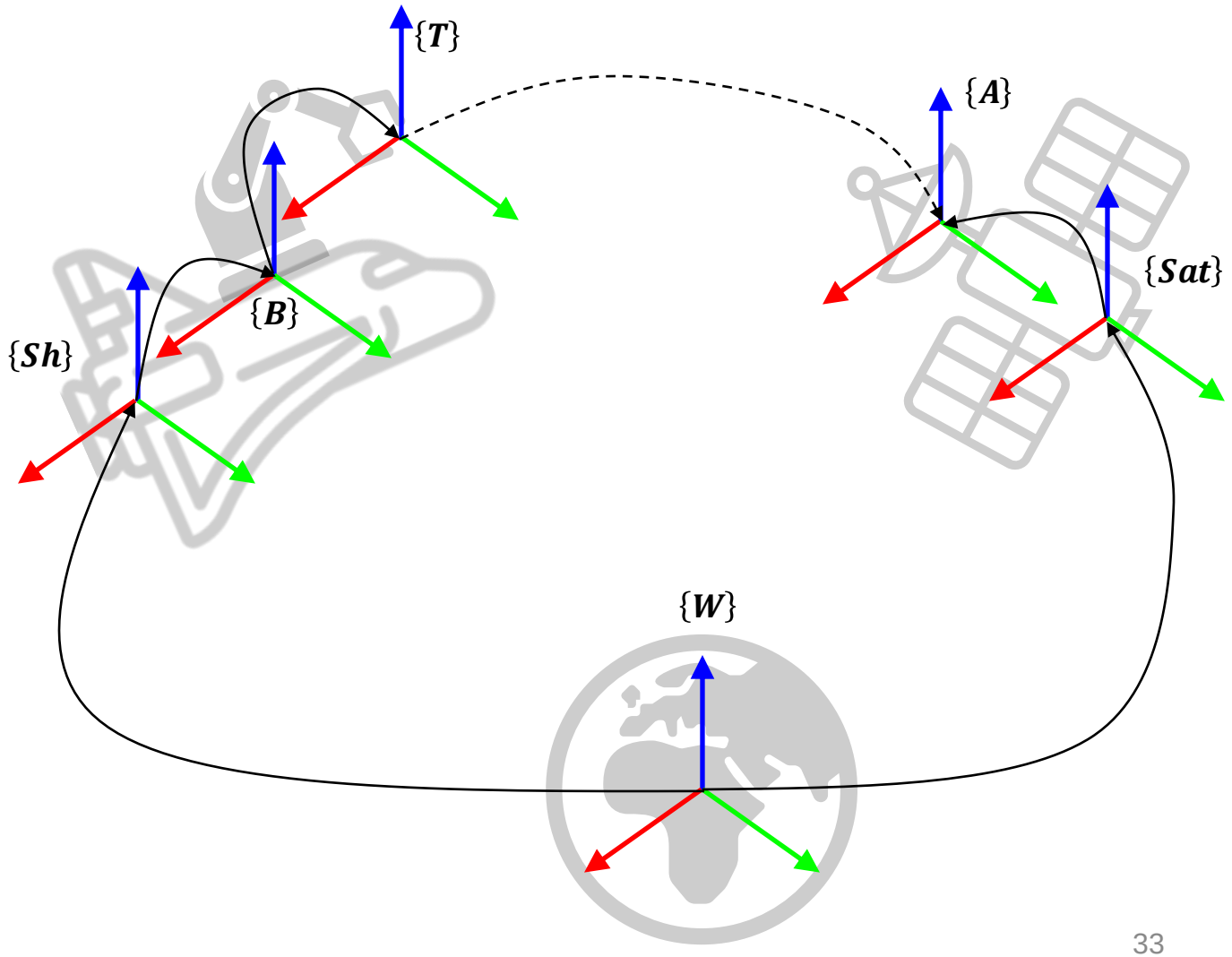
Wanted: ${}^{T}_{A}\boldsymbol{T}$



Example 2: Space Robotics

1. Write down the transform equation for the scenario.
2. Solve the equation for ${}^{T}_{A}\boldsymbol{T}$.
3. Which sensors could be used to make the various measurements?

Work individually.
(5-10 minutes)



What describes the transformation from {T} to {A}?

None of the above

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Example 2: Space Robotics – Solution

Think of it **mathematically**:

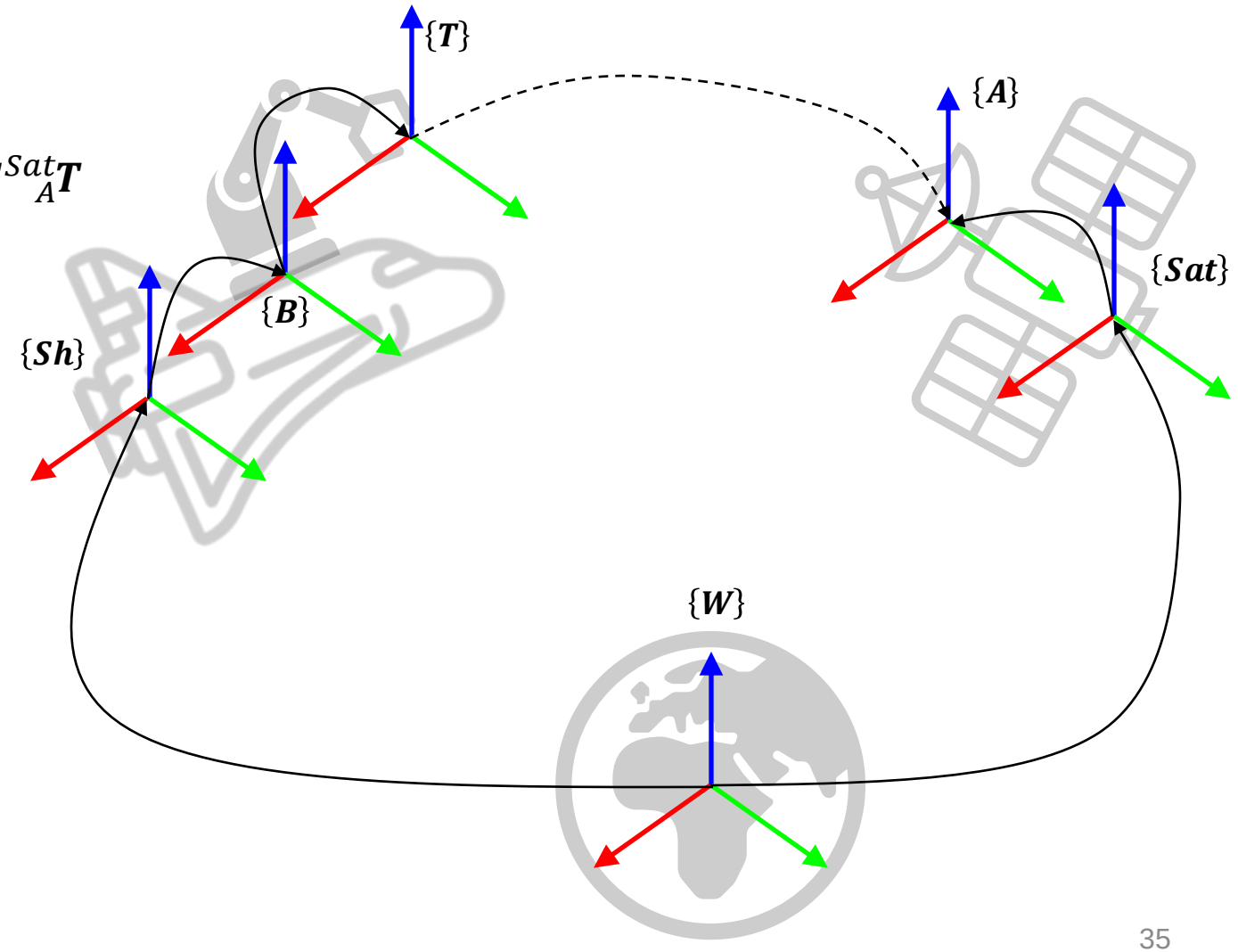
$$ {}^{W}_{Sh}\boldsymbol{T}\,{}^{Sh}_{B}\boldsymbol{T}\,{}^{B}_{T}\boldsymbol{T}\,{}^{T}_{A}\boldsymbol{T} = {}^{W}_{Sat}\boldsymbol{T}\,{}^{Sat}_{A}\boldsymbol{T} $$

$$ \left({}^{W}_{Sh}\boldsymbol{T}\,{}^{Sh}_{B}\boldsymbol{T}\,{}^{B}_{T}\boldsymbol{T}\right)^{-1}\,{}^{W}_{Sh}\boldsymbol{T}\,{}^{Sh}_{B}\boldsymbol{T}\,{}^{B}_{T}\boldsymbol{T}\,{}^{T}_{A}\boldsymbol{T} = \left({}^{W}_{Sh}\boldsymbol{T}\,{}^{Sh}_{B}\boldsymbol{T}\,{}^{B}_{T}\boldsymbol{T}\right)^{-1}\,{}^{W}_{Sat}\boldsymbol{T}\,{}^{Sat}_{A}\boldsymbol{T} $$

$$ {}^{T}_{A}\boldsymbol{T} = \left({}^{W}_{Sh}\boldsymbol{T}\,{}^{Sh}_{B}\boldsymbol{T}\,{}^{B}_{T}\boldsymbol{T}\right)^{-1}\,{}^{W}_{Sat}\boldsymbol{T}\,{}^{Sat}_{A}\boldsymbol{T} $$

Apply rule: $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$

$$ {}^{T}_{A}\boldsymbol{T} = {}^{B}_{T}\boldsymbol{T}^{-1}\,{}^{Sh}_{B}\boldsymbol{T}^{-1}\,{}^{W}_{Sh}\boldsymbol{T}^{-1}\,{}^{W}_{Sat}\boldsymbol{T}\,{}^{Sat}_{A}\boldsymbol{T} $$



Example 3: Calculating Tool Transformations

We want to **attach a Robotiq screwdriver** to a UR robot.

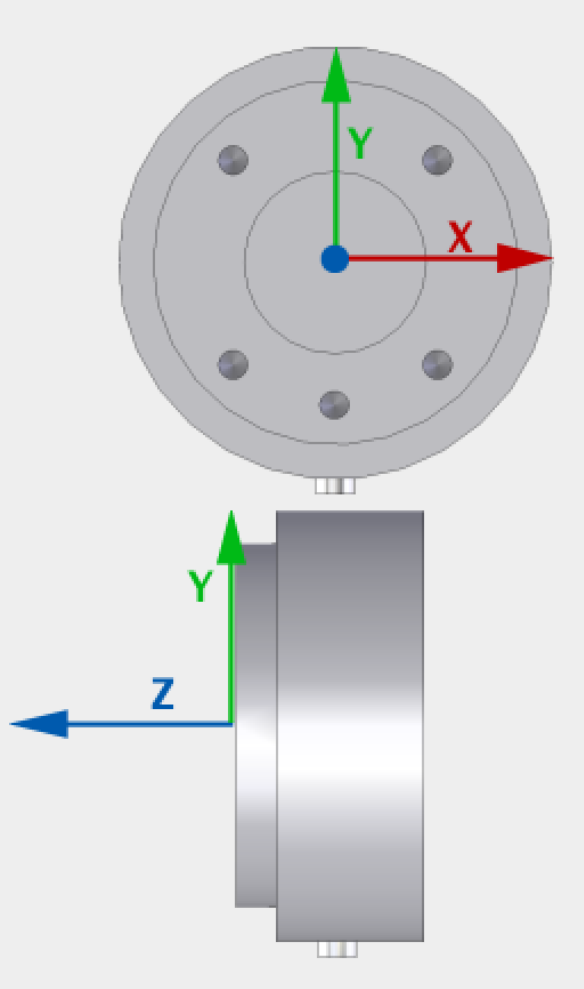
We need to calculate a **transformation** between the robot **tool flange** and the **tip of the screwdriver.**



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Example 3: Calculating Tool Transformations

The tool flange of a UR robot has the following orientation:



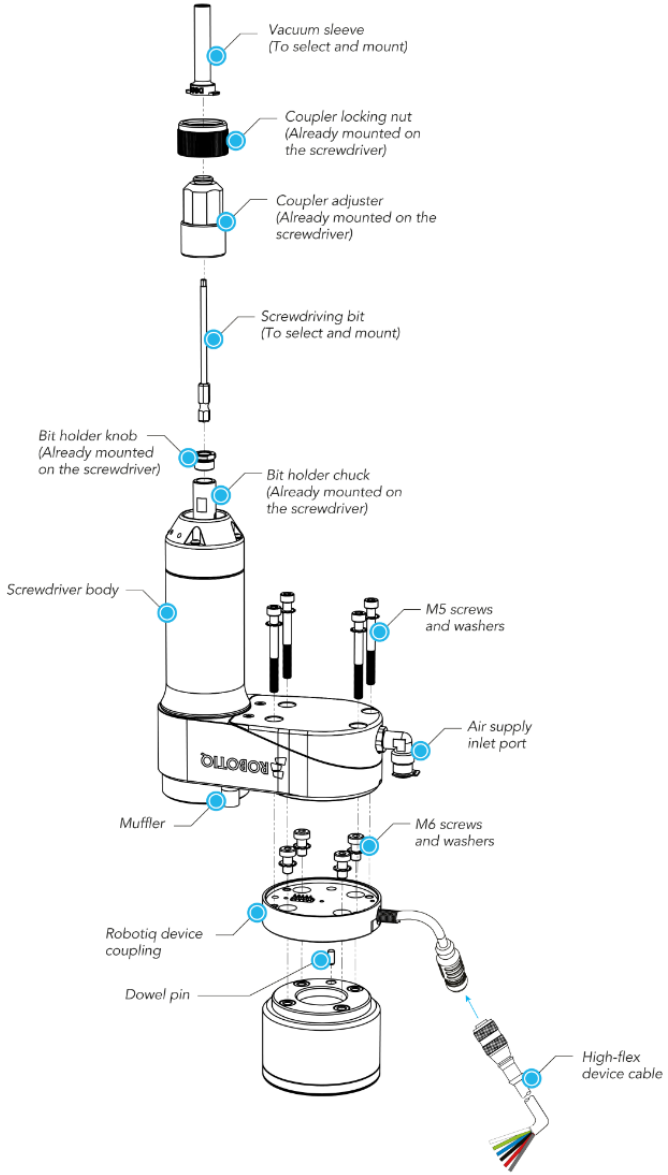


Fig. 3-1: Mechanical Installation of the Robotiq Screwdriver.

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Example 3: Calculating Tool Transformations

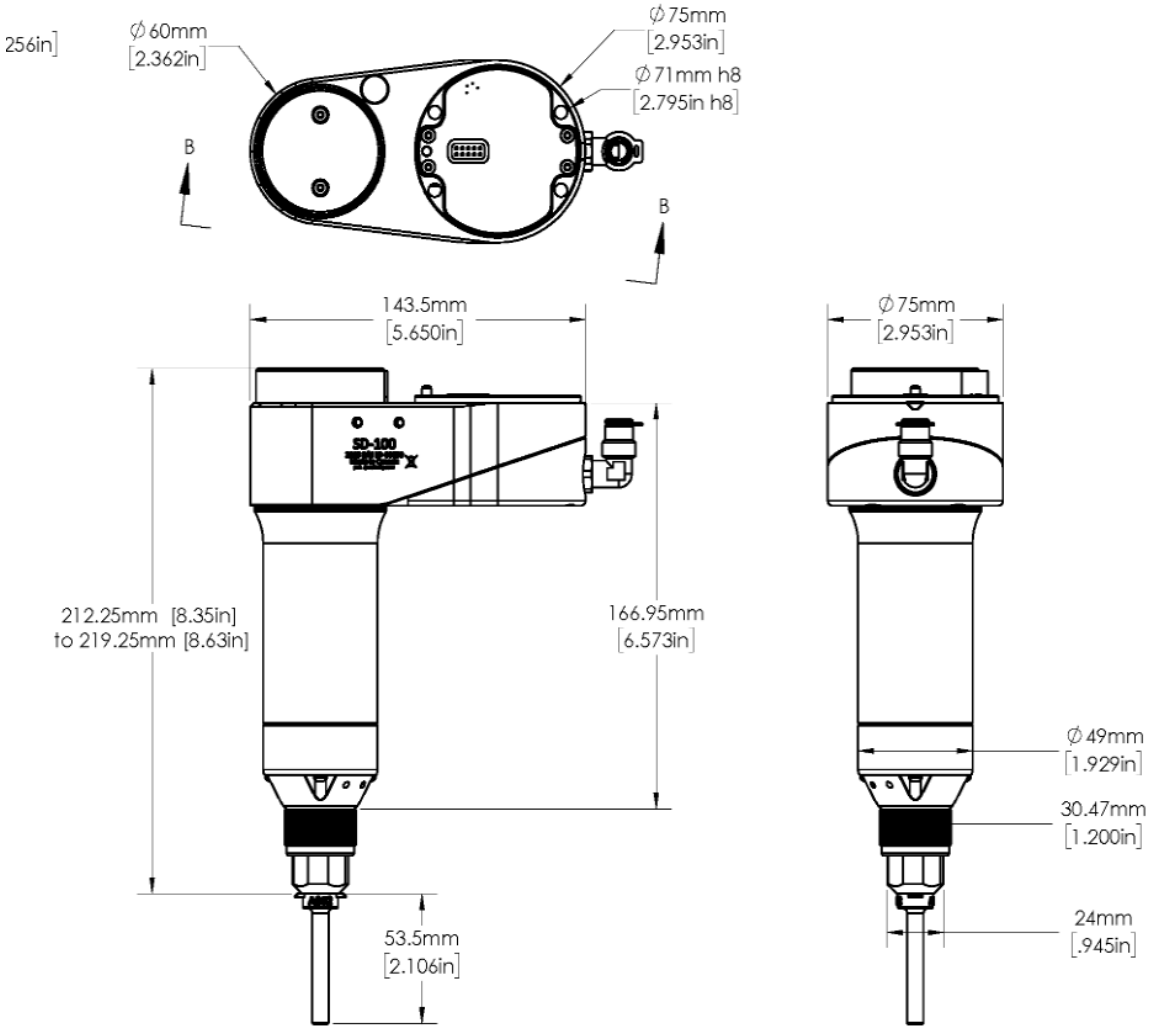


Fig. 7-1: Dimensions of the Screwdriver.

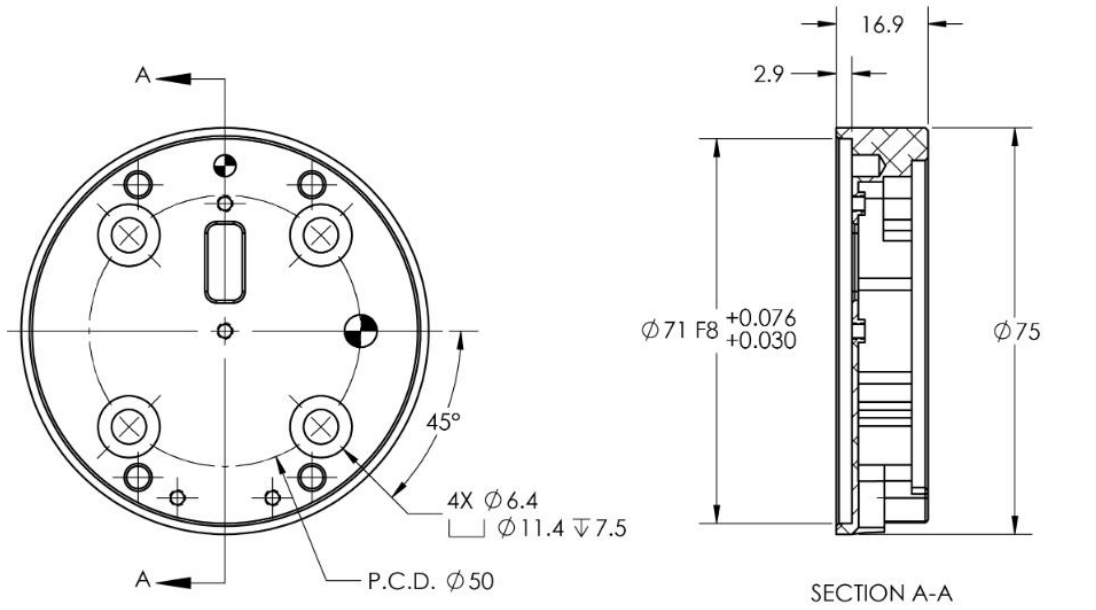


Fig. 7-3: Dimensions of the Robotiq Coupling for Installation of the Screwdriver.

Example 3: Calculating Tool Transformations

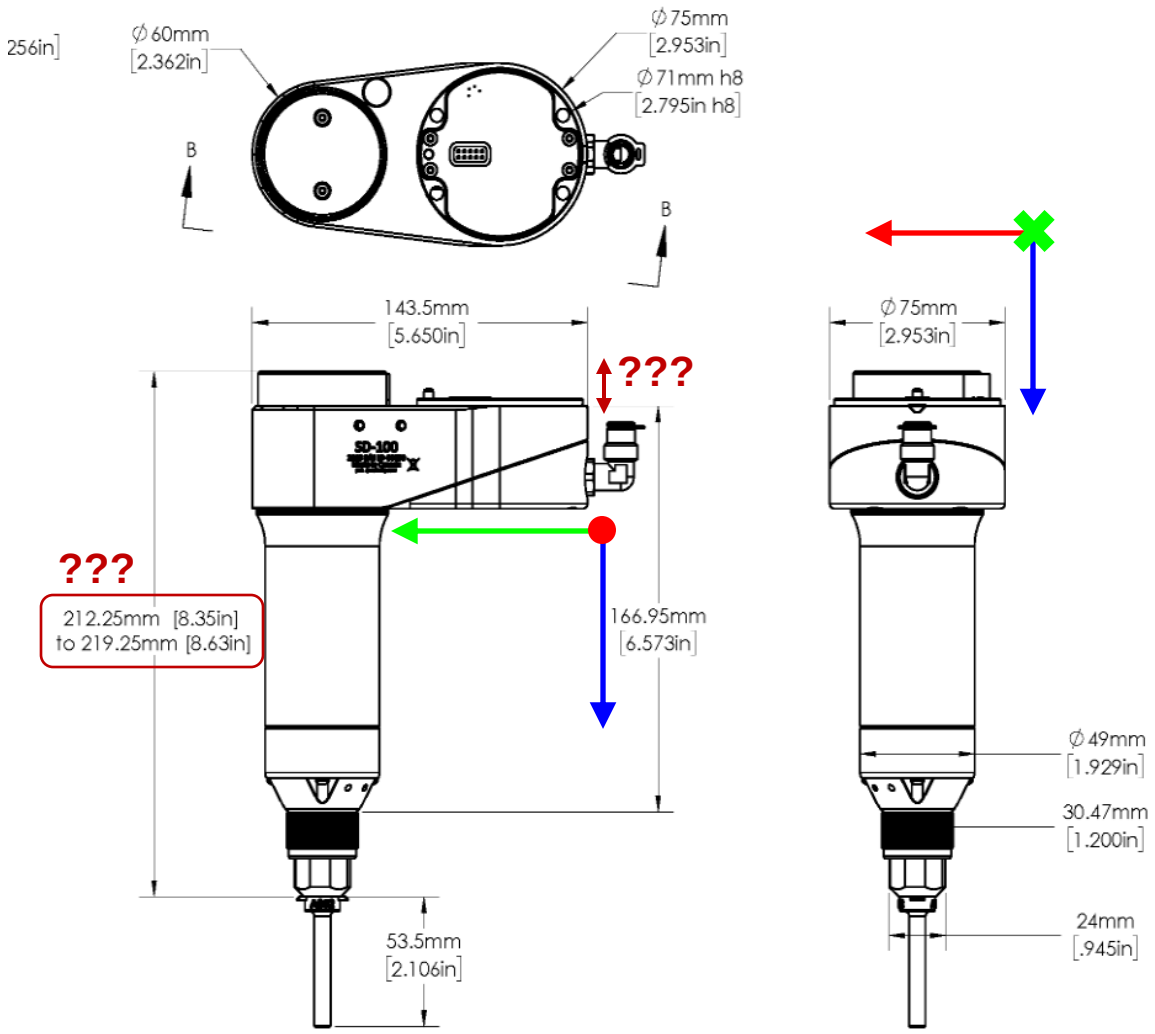


Fig. 7-1: Dimensions of the Screwdriver.

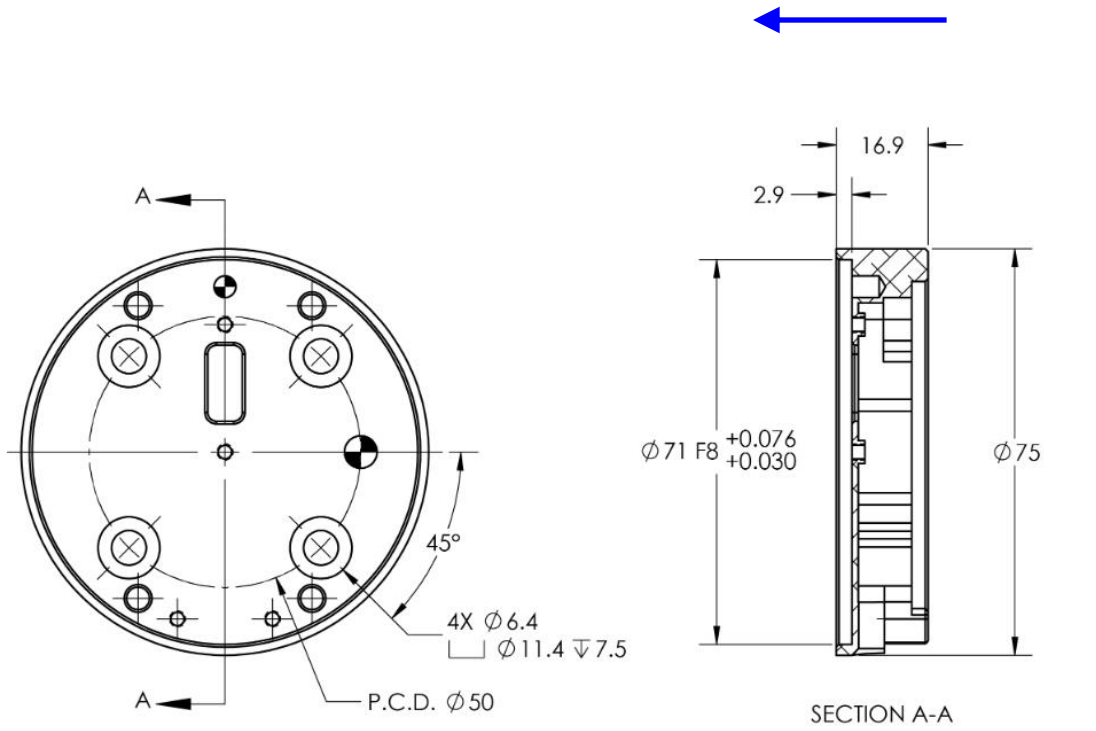


Fig. 7-3: Dimensions of the Robotiq Coupling for Installation of the Screwdriver.

Example 3: Calculating Tool Transformations

We have:

- **Rotation**: None → I_3
- **Translation:**
  - **x-axis**: $0\ \text{mm}$
  - **y-axis**: $143.55\ \text{mm} - \frac{60}{2}\text{mm} = 113.55\ \text{mm}$
  - **z-axis**: $219.25\ \text{mm} + 53.5\ \text{mm} + L_{\text{bit}} + L_{\text{screw}}$

With the information on the datasheet, this is underspecified.

For this example, let us just assume it is the longest measurement we are given.

In real life, you would measure it.

$$\boldsymbol{T}_{\textbf{screwdriver}}(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0.11355 \\ 0 & 0 & 1 & 0.27275 + L_{\text{bit}} + L_{\text{screw}} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Recap: What have we discussed today

- **Translational** and **Rotational Mappings**
- **Chains** of **Mappings**
- From **Euclidean** to **Projective Geometry** (and from **Cartesian** to **Homogenous Coordinates**)
- **Homogenous Transformation Matrices**:
  - **Shearing**
  - **Scaling**
  - **Computational considerations** (for example when taking the **inverse**)
- **Practical Applications**
  - **Mathematical** and **graphical approaches** to solutions

Take home message:

Homogenous Transformation Matrices allow us to **represent translation** and **rotation** together **in a single matrix** multiplication.

Many problems in robotics involve **solving** for the **missing transformation** in a chain. By **drawing** it, we can easily find a **graphical solution**.

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Thank you for today.

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